

9000 international standards to get their equipment calibrated in such a way that they maintain the same quality standards.

"Calibration should be traceable to national standard with unbroken chain of traceability, i.e., the equipment which is used as master should be calibrated with better accuracy master until the traceability reaches to National Physical Laboratory (NPL) or international standards," explains Desai.

In India, the highest levels of standards are maintained by NPL. "NPL gets their equipment calibrated from National Institute of Standards and Technology (NIST), Washington," shares Desai.

There are four Electronic Regional Test Laboratories (ERTLs) and government labs like Electronics Test & Development Centre (ETDC) and Centres for Electronics Test Engineering (CETE) that get their equipment calibrated from NPL and provide calibration services to others. Apart from this, there are private labs, either owned by the original equipment manufacturers (OEMs) or some third party, which provide calibration services and are accredited by National Accreditation Board for Testing and Calibration Laboratories (NABL). NABL provides laboratory accreditation as formal recognition of technical competence, while ISO certification is on quality system management. Thus most of test and measurement equipment manufacturers maintain their NABL-accredited calibration laboratory.

The hierarchy followed in calibrating one equipment from the master—which, in turn, is calibrated from higher levels of accuracies and standards—is maintained around the world. "You must test your equipment by a master that is at least a factor of four better than the equipment being calibrated," suggests Sulfikaruddin. If you are consulting a private laboratory, you must first ask for traceability of their standards and compare it with your equipment's accuracy. "Without

Calibration: Challenges to overcome

1. Identifying calibration needs and devoting time for calibrations
2. Lack of knowledge related to calibration cycle, calibration process and accredited laboratories
3. Lack of appreciation for the role of calibration
4. Calibrating the master/reference equipment and maintaining ideal environment for it
5. Calibration of equipment that don't have a set procedure and you have to test them with first principles
6. Lack of training for the calibration technician/engineer
7. Undetermined traceability

traceability, the calibration accuracy may be questionable," comments Bisenius.

Training the technician

All the measurement processes are accompanied by errors and uncertainties that cannot be eliminated. Sulfikaruddin says, "Calibration quantifies and controls the errors or uncertainties of the measurement processes to an acceptable level." This makes calibration a challenging job that must be handled by a trained technician or engineer.

According to Kela, the uncertainty of results is affected by 4 'M's—man, machine, method and material. Therefore utmost importance is given to train the people who perform calibration of equipment. They are also trained to use standard calibration methods and standard accessories to get accurate results.

Adjustment is not calibration

Most users are always confused between calibration and adjustment. It is important for the user to understand the difference between the two. When you are experimenting in a lab, and you do not see a 'zero' reading when the equipment is not connected, you make 'adjustments' to remove the offset. This is not calibration.

"Calibration is process of pa-

rameters verification of any equipment against a well-known standard. Calibration report is a data that shows what parameters of the equipment are within the manufacturer specification or what are out of specification. It does not include adjustment of parameters that are out of specification," explains Gaurav Tyagi, senior engineer-application and support at Aimil.

"Adjustment is a procedure in which parameters are adjusted within the specification range and usually done by the manufacturer or user," Tyagi adds. Although most companies maintain high levels of standards, they need to calibrate the equipment regularly.

Agreeing to this, Kela says, "Drift in accuracies over a period is unavoidable but can be detected in time, corrected mathematically and adjusted by calibration, i.e., by using a reference or a standard equipment. All calibration labs do not have capabilities of adjustments and fine-tuning."

Calibration cycle

For how long is your calibration certificate valid? Or how frequently you need to calibrate your equipment? This is a very subjective question. The answer depends on many factors like the type of the measuring equipment, usage of the equipment, man-handling and atmospheric changes that may influence the reading. In all the cases, these are variables.

"The calibration cycle of the equipment depends on the sensitivity of the equipment, environmental condition, usage, etc. Hence the calibration cycle varies from six months to 36 months depending upon the type of the equipment. Calibration cycles are generally recommended by OEMs," explains Sulfikaruddin.

"A regular user can get his equipment calibrated by having it checked by the OEM or sending it to an NABL-accredited laboratory with the details of the specifications of the product. Usually, it is required to be calibrated after an interval of twelve months from